

Boss Challenge — Paper 31 (Class 7)

Light | Transportation in Animals & Plants | Lines & Angles | Exponents & Powers

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. Light always travels from one place to another along a path that is:
(A) a circle
(B) a curved arc
(C) a zig-zag
(D) a straight line
2. The red fluid that carries food, oxygen and wastes around the human body is:
(A) saliva
(B) sweat
(C) blood
(D) bile
3. A pair of angles whose measures together add up to 90° are called:
(A) supplementary angles
(B) complementary angles
(C) adjacent angles
(D) vertically opposite angles
4. In the expression 5^3 , the number 5 is called the:
(A) factor count
(B) base
(C) product
(D) exponent
5. A mirror with a flat, smooth reflecting surface is called a:
(A) convex mirror
(B) plane mirror
(C) concave mirror
(D) lens
6. The liquid part of blood, in which the cells float, is called:
(A) haemoglobin
(B) lymph
(C) plasma
(D) platelets
7. Two angles whose measures add up to 180° are called:
(A) equal angles
(B) supplementary angles
(C) right angles
(D) complementary angles

8. In the expression 2^6 , the small raised number 6 is the:
- (A) quotient
 - (B) exponent
 - (C) base
 - (D) sum
9. The image you see of yourself in a plane mirror is always:
- (A) erect but smaller
 - (B) inverted but same size
 - (C) erect and the same size
 - (D) upside down and bigger
10. The red colour of blood comes from a substance in red blood cells called:
- (A) haemoglobin
 - (B) plasma
 - (C) platelets
 - (D) insulin
11. The complement of a 35° angle is:
- (A) 145°
 - (B) 65°
 - (C) 35°
 - (D) 55°
12. Worked out in full, the value of 3^4 is:
- (A) 27
 - (B) 64
 - (C) 12
 - (D) 81
13. In a plane mirror your right hand appears to be the image's left hand. This swap is called:
- (A) refraction
 - (B) dispersion
 - (C) lateral inversion
 - (D) magnification
14. The blood cells that fight germs and disease are the:
- (A) platelets
 - (B) nerve cells
 - (C) red blood cells
 - (D) white blood cells
15. For an angle measuring 110° , its supplement works out to:
- (A) 110°
 - (B) 20°
 - (C) 80°
 - (D) 70°

16. The value of 10^5 is:

- (A) 1,00,000
- (B) 50
- (C) 10,000
- (D) 1,000

17. The image formed by a plane mirror cannot be caught on a screen, so it is called a:

- (A) reflection-free image
- (B) virtual image
- (C) shadow
- (D) real image

18. When you cut your finger, the bleeding stops because tiny blood cell pieces called platelets help the blood to:

- (A) freeze
- (B) boil
- (C) clot
- (D) evaporate

19. When two straight lines cross, the angles opposite each other at the crossing are always:

- (A) equal
- (B) supplementary
- (C) complementary
- (D) always 90°

20. Using laws of exponents, $2^3 \times 2^9$ equals:

- (A) 4^9
- (B) 2^9
- (C) 2^{12}
- (D) 2^1

21. A mirror that curves inward, like the inside of a spoon's bowl, is a:

- (A) plane mirror
- (B) convex mirror
- (C) plain glass
- (D) concave mirror

22. The muscular organ that pumps blood all around the body is the:

- (A) kidney
- (B) heart
- (C) lung
- (D) liver

23. Two angles that share a common arm and a common vertex but do not overlap are:
- (A) adjacent angles
 - (B) alternate angles
 - (C) complementary angles
 - (D) vertically opposite angles
24. Using laws of exponents, $5^4 / 5^2$ equals:
- (A) 1
 - (B) 5
 - (C) 5^3
 - (D) 5^4
25. A concave mirror can gather light from a far object and form an image that is:
- (A) real and can be caught on a screen
 - (B) coloured like a rainbow
 - (C) virtual and never on a screen
 - (D) always the same size as you
26. Blood vessels that carry blood away from the heart to the body are the:
- (A) capillaries
 - (B) arteries
 - (C) nerves
 - (D) veins
27. Two adjacent angles whose outer arms form a straight line add up to:
- (A) 180°
 - (B) 360°
 - (C) 270°
 - (D) 90°
28. The value of any non-zero number raised to the power 0, such as 7^0 , is:
- (A) 1
 - (B) 7
 - (C) 70
 - (D) 0
29. A mirror that bulges outward, like the back of a spoon, always gives an image that is:
- (A) real and on a screen
 - (B) the same size always
 - (C) erect and smaller (diminished)
 - (D) inverted and larger
30. Blood vessels that bring blood back to the heart are the:
- (A) windpipe
 - (B) arteries
 - (C) capillaries
 - (D) veins

31. In a linear pair, if one angle is 125° , the other angle is:
- (A) 125°
 - (B) 55°
 - (C) 65°
 - (D) 35°
32. Using the power-of-a-power law, $(2^3)^2$ equals:
- (A) 2■
 - (B) 2■
 - (C) 4■
 - (D) 2■
33. Side mirrors on cars and scooters are convex because they:
- (A) show a wider area of traffic behind
 - (B) magnify each vehicle
 - (C) only work at night
 - (D) flip the road upside down
34. The extremely thin vessels where food and oxygen pass from blood into the cells are the:
- (A) capillaries
 - (B) arteries
 - (C) veins
 - (D) ureters
35. A line that crosses two other lines is called a:
- (A) transversal
 - (B) diameter
 - (C) ray
 - (D) perpendicular
36. Expressed in standard (scientific) form, the number 47000 is:
- (A) 0.47×10 ■
 - (B) 47×10^3
 - (C) 4.7×10^3
 - (D) 4.7×10 ■
37. A piece of glass that bends light passing through it to form images is a:
- (A) prism
 - (B) lens
 - (C) screen
 - (D) plane mirror
38. The throbbing you feel at your wrist, caused by blood being pushed through an artery, is the:
- (A) reflex
 - (B) pulse
 - (C) breath
 - (D) clot

39. If a transversal crosses two parallel lines, then each pair of corresponding angles is:

- (A) equal
- (B) supplementary
- (C) always right angles
- (D) complementary

40. Written as a power of 2, the number 32 is:

- (A) 5^2
- (B) 2^5
- (C) 2^6
- (D) 2^7

41. A lens that is fatter in the middle than at its edges is a:

- (A) concave lens
- (B) plane mirror
- (C) convex lens
- (D) prism

42. The waste cleaning organs that filter the blood and make urine are the:

- (A) lungs
- (B) heart
- (C) kidneys
- (D) stomach

43. When a transversal cuts two parallel lines, a pair of alternate interior angles is:

- (A) complementary
- (B) unequal always
- (C) supplementary
- (D) equal

44. Expanded as repeated multiplication, 4^3 means:

- (A) $3 \times 3 \times 3 \times 3$
- (B) 4×3
- (C) $4 \times 4 \times 4$
- (D) $4 + 4 + 4$

45. A convex lens used to read tiny print acts as a:

- (A) a shrinking glass
- (B) a mirror
- (C) magnifying glass
- (D) a torch

46. The tube that carries urine from a kidney down to the urinary bladder is the:

- (A) ureter
- (B) urethra
- (C) artery
- (D) windpipe

47. When a transversal cuts two parallel lines, co-interior (same-side interior) angles add up to:

- (A) 360°
- (B) 90°
- (C) equal to each other
- (D) 180°

48. FUSION (Transportation): a resting heart beats about 72 times a minute. The number of beats in one minute, written in standard form to 1 decimal place, is closest to:

- (A) 7.2×10^1
- (B) 7.2×10^2
- (C) 72×10 ■
- (D) 0.72×10^2

49. A lens that is thinner in the middle than at the edges always makes objects look:

- (A) upside down
- (B) larger
- (C) smaller
- (D) coloured

50. Sweat helps the body by removing some water and salts and also by:

- (A) adding salt to the blood
- (B) warming the body up
- (C) making the skin waterproof
- (D) cooling the body as it evaporates

51. An angle that is more than 90° but less than 180° is called:

- (A) a reflex angle
- (B) an obtuse angle
- (C) a right angle
- (D) an acute angle

52. FUSION (Transportation): a heart beating 72 times a minute beats about $72 \times 60 = 4320$ times an hour. In standard form, 4320 is:

- (A) 4.32×10^2
- (B) 43.2×10^2
- (C) 4.32×10^3
- (D) 4.32×10 ■

53. White sunlight is split into its colours when it passes through a:

- (A) convex lens
- (B) sheet of paper
- (C) plane mirror
- (D) prism

54. In plants, water and dissolved minerals are carried upward from the roots through tubes called:

- (A) phloem
- (B) veins
- (C) xylem
- (D) stomata

55. An angle greater than 180° but less than 360° is called:

- (A) a reflex angle
- (B) a complete angle
- (C) an obtuse angle
- (D) a straight angle

56. FUSION (Transportation): one drop of blood holds roughly 5,000,000 red blood cells. Written in standard form, this count is:

- (A) $5 \times 10^{\blacksquare}$
- (B) $50 \times 10^{\blacksquare}$
- (C) $5 \times 10^{\blacksquare}$
- (D) $5 \times 10^{\blacksquare}$

57. The band of seven colours obtained when white light is split is called the:

- (A) reflection
- (B) image
- (C) shadow
- (D) spectrum

58. The tubes that carry food made in the leaves to all other parts of a plant are the:

- (A) xylem
- (B) phloem
- (C) arteries
- (D) roots

59. FUSION (Light): a ray of light strikes a plane mirror making a 30° angle with the mirror surface. The angle of incidence, measured from the normal, is:

- (A) 120°
- (B) 60°
- (C) 90°
- (D) 30°

60. FUSION (Transportation): an adult body holds about 5 litres of blood. Written in millilitres (1 litre = 1000 mL), 5 litres in standard form is:

- (A) 5×10^3 mL
- (B) 50×10^2 mL
- (C) $5 \times 10^{\blacksquare}$ mL
- (D) 5×10^2 mL

61. A rainbow in the sky is nature's way of splitting sunlight using tiny:
- (A) water drops
 - (B) ice cubes
 - (C) dust grains
 - (D) clouds alone
62. The continuous loss of water vapour mainly through the tiny pores of leaves is called:
- (A) transpiration
 - (B) condensation
 - (C) germination
 - (D) respiration
63. FUSION (Light): the law of reflection says the angle of incidence equals the angle of reflection. If a ray hits a mirror at an angle of incidence of 40° , the angle of reflection is:
- (A) 40°
 - (B) 20°
 - (C) 50°
 - (D) 80°
64. The number 1,00,00,000 (one crore) is written as a power of ten as:
- (A) 10^8
 - (B) 10^7
 - (C) 7^1
 - (D) 10^9
65. We can see a non-luminous object such as a book only because it:
- (A) makes its own light
 - (B) bends light like a lens
 - (C) reflects light into our eyes
 - (D) absorbs all the light
66. The tiny pores on the underside of leaves through which water vapour and gases pass are the:
- (A) xylem
 - (B) stomata
 - (C) roots
 - (D) veins
67. FUSION (Light): a light ray hits a plane mirror straight on, along the normal (angle of incidence 0°). The total angle between the incoming and reflected rays is:
- (A) 180°
 - (B) 60°
 - (C) 90°
 - (D) 0°

68. Using laws of exponents, $3^3 / 3^3$ equals:

- (A) 1
- (B) 3^3 left unsimplified
- (C) 0
- (D) 3

69. Bouncing of light back from a surface like a mirror is called:

- (A) absorption
- (B) dispersion
- (C) refraction
- (D) reflection

70. Transpiration from the leaves helps a plant by creating a pull that:

- (A) pushes food down to the roots
- (B) cools the soil around the roots
- (C) draws water up from the roots
- (D) makes the roots grow longer

71. FUSION (Light): for a ray reflecting off a plane mirror with angle of incidence 35° , the angle between the incident ray and the reflected ray equals:

- (A) 35°
- (B) 70°
- (C) 90°
- (D) 17.5°

72. Comparing the two powers, which is greater - 2^5 or 5^2 ?

- (A) 5^2 , because the base is bigger
- (B) 5^2 , because it equals 50
- (C) They are equal
- (D) 2^5 , because it equals 32 while 5^2 equals 25

73. The image in a plane mirror appears to be as far behind the mirror as the object is:

- (A) inside the glass
- (B) twice as far behind
- (C) in front of it
- (D) above the mirror

74. Single-celled animals like Amoeba do not need a transport system because:

- (A) they never need oxygen
- (B) they store all food forever
- (C) substances reach every part by simple diffusion
- (D) they have tiny hearts

75. An angle exactly equal to 90° is a:

- (A) acute angle
- (B) reflex angle
- (C) right angle
- (D) straight angle

76. Writing 64 as a power of 4 gives:

- (A) 4^4
- (B) 4^3
- (C) 6^4
- (D) 4^2

77. A spoon's inside (bowl) makes your face look upside down when held far away because the inner surface acts like a:

- (A) concave mirror
- (B) lens
- (C) plane mirror
- (D) convex mirror

78. The number of times the heart beats in one minute is called the:

- (A) blood pressure
- (B) breathing depth
- (C) heartbeat rate (pulse rate)
- (D) clotting time

79. If two angles are equal and also supplementary, each one must be:

- (A) 60°
- (B) 90°
- (C) 45°
- (D) 180°

80. Using laws of exponents, $(a \times b)^2$ equals:

- (A) $a \times b^2$
- (B) $a^2 \times b^2$
- (C) $2ab$
- (D) $a^2 + b^2$

81. A real image, unlike a virtual one, is always:

- (A) formed behind the mirror
- (B) able to be caught on a screen
- (C) only seen in plane mirrors
- (D) always upright

82. Compared with arteries, the walls of veins are usually:

- (A) thinner, and veins have valves
- (B) made of bone
- (C) thicker, with no valves
- (D) exactly the same as arteries

83. The angle made by two arms that point in exactly opposite directions along a straight line is a:

- (A) full angle of 360°
- (B) zero angle of 0°
- (C) straight angle of 180°
- (D) right angle of 90°

84. The value of 1 raised to any power, such as $1^{\blacksquare}$, is:

- (A) 0
- (B) 19
- (C) 9
- (D) 1

85. If you move toward a plane mirror, your image moves:

- (A) away from you
- (B) stays exactly still
- (C) grows much larger
- (D) toward you by the same amount

86. Tall trees can raise water many metres to their topmost leaves mainly because of root pressure and:

- (A) wind blowing it upward
- (B) the heart pumping it up
- (C) sunlight pushing the water
- (D) the pull created by transpiration

87. Vertically opposite angles are formed when:

- (A) two straight lines intersect
- (B) a single ray is drawn
- (C) two lines are parallel
- (D) two lines never meet

88. Written in standard form, the number 0.0006 is:

- (A) $6 \times 10^{\blacksquare}$
- (B) $0.6 \times 10^{\blacksquare^3}$
- (C) $6 \times 10^{\blacksquare\blacksquare}$
- (D) $6 \times 10^{\blacksquare^3}$

89. A dentist uses a concave mirror to look at teeth because it can give an:
- (A) colour-split image
 - (B) smaller image of the tooth
 - (C) enlarged, erect image when held close
 - (D) upside-down view for comfort
90. Roots take in water from the soil through tiny hair-like outgrowths called:
- (A) root hairs
 - (B) petals
 - (C) stomata
 - (D) phloem
91. If two parallel lines are cut by a transversal and one corresponding angle is 75° , then the matching corresponding angle is:
- (A) 105°
 - (B) 90°
 - (C) 75°
 - (D) 15°
92. Using laws of exponents, $7^m \times 7^n$ equals:
- (A) 0
 - (B) 7^m
 - (C) 7^n
 - (D) 7^{m+n}
93. Headlights of a car use a concave mirror behind the bulb to:
- (A) dim the bulb to save power
 - (B) scatter light in all directions
 - (C) split the light into colours
 - (D) throw the light forward as a strong beam
94. Excretion is the removal from the body of:
- (A) undigested food only
 - (B) harmful waste substances made by the body
 - (C) water that the body needs
 - (D) useful food and oxygen
95. Two lines in a plane that never meet, no matter how far they are extended, are called:
- (A) intersecting lines
 - (B) perpendicular lines
 - (C) concurrent lines
 - (D) parallel lines

96. The value of 2^{10} is:

- (A) 100
- (B) 1024
- (C) 2048
- (D) 512

97. Two plane mirrors facing each other in a kaleidoscope create:

- (A) no image at all
- (B) a real image on a screen
- (C) a single small image
- (D) many repeated images of the coloured bits

98. Into how many chambers is the human heart divided?

- (A) two
- (B) one
- (C) four
- (D) three

99. An angle that measures less than 90° is called:

- (A) an obtuse angle
- (B) a reflex angle
- (C) an acute angle
- (D) a right angle

100. The number 1000 written as a power of ten is:

- (A) 10^3
- (B) 10^2
- (C) 10^4
- (D) 3^{10}